

VERIFICATION OF KIRCHOFF'S VOLTAGE LAW AND KIRCHOFF'S CURRENT LAW

Aim:

To verify Kirchhoff's voltage law for loop 1 and Kirchhoff's law in node A for the following circuit.

APPARATUS REQUIRED:

Laptop with Proteus Software

Theory:

According to Kirchhoff's voltage law sum of potential rises are equal to sum of potential drop in a loop.

In loop 1 By KVL:

$$V = V_1 + V_2 + V_3$$

$$100 = IR_1 + IR_2 + IR_3$$

$$R_{eq} = R_1 + R_2 + R_3$$

$$R_{eq} = 30 + 50 + 100$$

$$R_{eq} = 180 \Omega$$

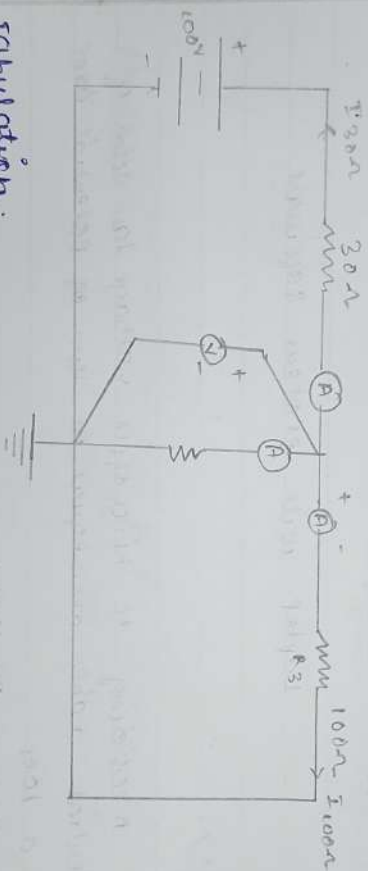
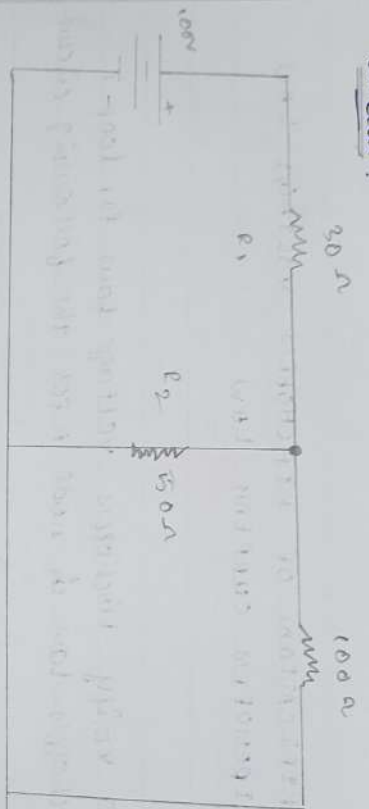
$$I = \frac{V}{R_{eq}} = \frac{100}{180} = 0.556 A$$

$$V_1 = V_{30\Omega} = IR_1 = 0.556 \times 30 = 16.68 V$$

$$V_2 = V_{50\Omega} = IR_2 = 0.556 \times 50 = 27.8 V$$

$$V_3 = V_{100\Omega} = IR_3 = 0.556 \times 100 = 55.6 V$$

Circuit:



Calculation:

KCL	$I_{30\Omega}$	$I_{50\Omega}$	$I_{100\Omega}$	V_R
Theoretical	-1.48 A	1.85 A	0.56 A	52.63 V
Practical	1.58 A	1.05 A	0.53 A	52.6 V

Substitute V_1, V_2 and V_3 values in (1)

$$100 = 16.68 + 24.8 = 55.6$$

$$100V = 100V$$

Hence proved.

KIRCHHOFF'S CURRENT LAW:

In a junction, incoming current is equal to outgoing current.

At node A

$$I_{30\Omega} + I_{50\Omega} + I_{100\Omega} = 0$$

$$\frac{V - 100}{30} + \frac{V}{50} + \frac{V}{100} = 0$$

$$V \left[\frac{1}{30} + \frac{1}{50} + \frac{1}{100} \right] - \frac{100}{30} = 0$$

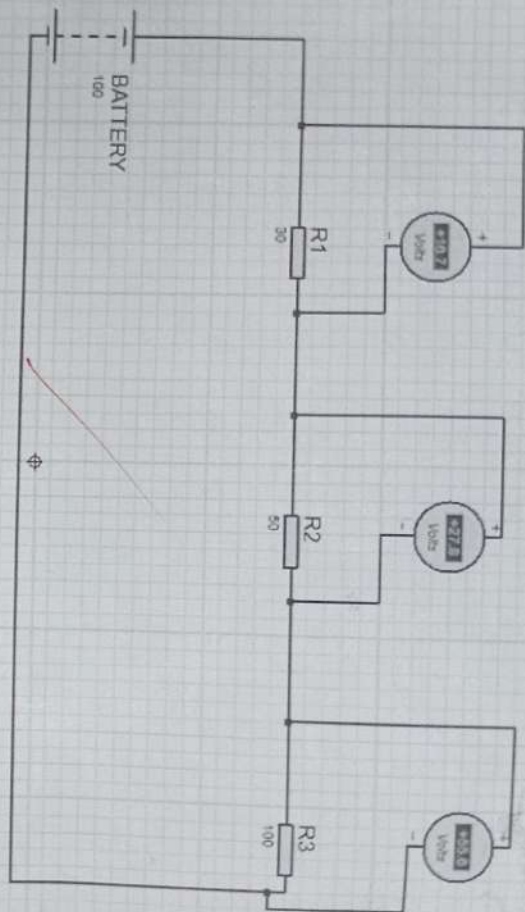
$$V [0.0633] - 3.333 = 0$$

$$V = \frac{3.333}{0.0633}$$

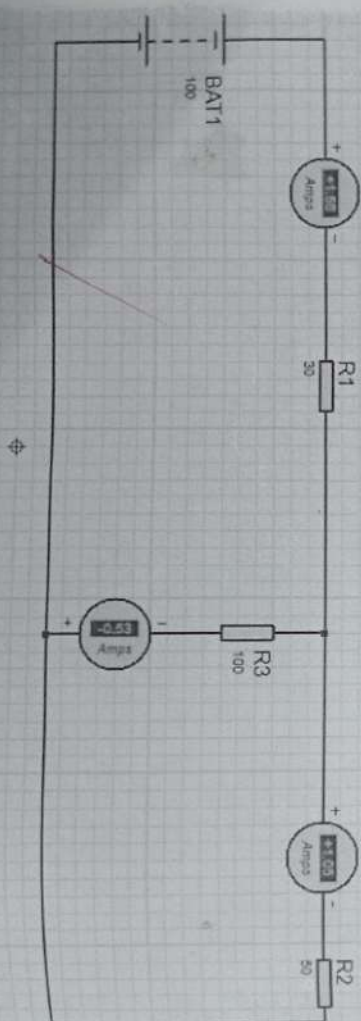
$$V = 52.63V$$

$$I_{30\Omega} = \frac{V - 100}{30} = \frac{52.63 - 100}{30} = -1.58A$$

$$I_{50\Omega} = \frac{V}{50} = \frac{52.63}{50} = 1.05A$$



Expt. No.



$$I_{100\Omega} = \frac{V}{100} = \frac{52.63}{100}$$

$$= 0.53A$$

At node A:

$$I_{30\Omega} + I_{50\Omega} + I_{100\Omega} = 0$$

$$-1.58 + 1.05 + 0.53 = 0$$

Hence proved.

Result:

Thus, the Kirchhoff's voltage law and Kirchhoff's current law has been verified successfully.